

Shayan Shakeri

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PROFESSIONAL SUMMARY

AI Research Engineer at BMO's AI Centre of Excellence. Built BMO Wealth's multi-agent data layer from the ground up, scaled it, and own it today: an orchestration harness giving agents graph, analysis, and search tools over client data. Built the eval pipeline that caught a production GenAI tool serving \$200B+ AUM downplaying investment risk. Took a proprietary subcutaneous neurostimulator from idea to production device at Epineuron. Co-founder of Merlyn Labs, researching VLAs and designing games for agent evaluations; M.Eng in AI & Robotics, University of Toronto, expected December 2026.

WORK EXPERIENCE

BMO AI Centre of Excellence

Sep 2025 - Present

AI Research Engineer

Toronto

- Built, scaled, and own BMO Wealth's multi-agent data layer: an orchestration harness with graph, analysis, and search tools
- One prompt found, sorted, and screened nearly 30,000 clients for defined-benefit plans in 3 hours, previously days of manual work
- Built a deterministic agent eval pipeline using hundreds of synthesized inputs to detect misaligned outputs at scale
- Uncovered systematic bias to downplay investment risk in a GenAI tool serving \$200B+ AUM wealth management
- Building internal infrastructure that provisions LLM agents from a written role and scope definition
- Designing evals for LLM agents that retrieve and reason over commercial banking and insurance policy

Merlyn Labs

Aug 2025 - Present

AI Researcher & Co-Founder

Toronto

- Showed $\pi 0.5$ generalization failures on LIBERO-PRO are recipe-induced overfitting; proposed an alternative baseline
- Conservative finetuning doubled position-swap success (21% to 42%), showing spatial generalization over memorization
- Developing VLM judges that score rollouts into dense, context-dependent RL rewards that are difficult to game

Epineuron Technologies

May 2021 - Aug 2022

Biomedical & Electrical Engineering Co-op

Mississauga

- Owned a proprietary subcutaneous neurostimulator from first idea to production device at a clinical-stage startup
- Designed and assembled the PCBs and wrote the IEC 60601-1 / ISO 13485 validation protocols for clinical test runs
- Part of the team behind PeriPulse, an FDA Breakthrough-designated device now in multinational clinical trials
- Extended battery life 900% through oscilloscope and power-analyzer testing of the device's power draw
- Modeled electromagnetic field penetration into peripheral nerves in COMSOL; the results set the electrode diameter

RESEARCH & PROJECTS

Stanford BEHAVIOR-1K Challenge, 8th Place, Standard Track

Sep - Nov 2025

- Identified proprioceptive collapse as a critical VLA failure mode; 60% masking improved task success by up to 48%
- Trained on 10,000+ demonstrations covering 22 of 50 scored task types; published a technical report and a LessWrong analysis

BardSong, AI-Powered D&D Storytelling Tool

Jun 2025 - Present

- Built end-to-end pipeline: Groq speech-to-text, Gemini scene extraction, image generation, narrated video
- Fine-tuned LLM on D&D sourcebooks for improved narrative coherence; currently in closed alpha with 23 DMs

VP Technical, McMaster Medical Design Team

Apr 2018 - Apr 2023

- Led development of open-source motorized prosthetic for lower-arm amputees that anyone could assemble for <\$300

EDUCATION

M.Eng in AI & Robotics University of Toronto

Sep 2024 - Dec 2026, expected

Coursework: Deep Learning, Reinforcement Learning, AI Applications in Robotics, Healthcare Robotics

B.Eng, Engineering Physics Co-op, Dean's Honour List McMaster University

Sep 2018 - Jun 2023

SKILLS & INTERESTS

ML & AI: Agents, Agent Evaluation, Alignment Evaluation, LLM Fine-tuning, Python, C/C++, PyTorch, JAX, RL, Imitation Learning

Robotics & Hardware: PCB Design, 3D Printing, VLA Models, OmniGibson, MuJoCo, Flow Matching, Sim2Real, ROS2

Interests: Literature, Dungeons & Dragons, Volleyball, Twilight Imperium, Piano